Paper 1 (MCQ)

1	**A**	6	**C**	11	**B**	16	**D**	21	**B**	26	**D**	31	**A**	36	**A**
2	**C**	7	**D**	12	**A**	17	**A**	22	**C**	27	**B**	32	**C**	37	**D**
3	**C**	8	**B**	13	**D**	18	**C**	23	**D**	28	**A**	33	**A**	38	**C**
4	**D**	9	**A**	14	**C**	19	**A**	24	**C**	29	**C**	34	**A**	39	**D**
5	**B**	10	**B**	15	**A**	20	**C**	25	**A**	30	**C**	35	**D**	40	**A**

Paper 2 (CORE)
Section A: Structured Answers

QUESTION 1
With reference to **Figs. 1.1** & **1.2**,
(a) state one similarity and one difference in the structure of glycogen and triglyceride. [2]

Similarity:
- both are made up of **carbon, oxygen and hydrogen**
- both are **covalently bonded**
- both are **formed through condensation reactions**
- both have **high % of hydrocarbons (C-H)**

Difference:

Glycogen	Triglyceride
Monomers are made up of **glucose**	Monomers are made up of **glycerol & fatty acid chains**
Bonds linking up the monomers are **glycosidic linkages**	Bonds linking up the monomers are **ester linkages**

(b) Describe how the hydrogen released during glycolysis and the Krebs cycle results in the production of ATP. [2]
- Hydrogen released are used **to reduce coenzymes NAD^+ & FAD^+**
- The NADH and $FADH_2$ will be **reoxidized** through transfer of their electrons and protons **to the electron transport chain** at the inner membrane of the mitochondrion.
- **Protons** are pumped from the mitochondrial matrix **into the intermembranal space generating an electrochemical gradient**
- The re-entry of the protons to into the matrix provides the energy to **generate ATP** at the **ATP synthase / ATP synthetase / ATPase**
- Ref to **chemiosmosis / oxidative phosphorylation**

(c)(i) Using the information from Fig. **1.3**, calculate the number of ATP molecules produced from the complete oxidation of a 16-carbon fatty acid chain. Show your answer in the space provided below. [2]
16-C fatty acid = **8 x 2-C chains,** where one 2-C chain produces **12 ATP molecules.** Eight 2-C chains produce **12 x 8** = **96 ATP molecules**

(c)(ii) Suggest why fatty acids can only be respired under aerobic conditions. [2]
Fatty acids enter aerobic respiration by being **converted to acetyl coA and enters the Krebs cycle, bypassing glycolysis.** In order to be completed oxidized, **oxygen is needed as final electron acceptor in the process of oxidative phosphorylation**
Ref to **link reaction**

(d)(i) Explain two important principles of homeostasis that are illustrated in **Fig. 1.4**. [3]
- **Self regulation** (OWTTE)
  To maintain a constant internal environment despite changes in the external environment, **the normal blood glucose level is** the set point at approximately 0.9g dm^{-3}. Deviation from the norm / set point is detected by the pancreatic islet cells. **Deviation acts as a stimulus that trigger either insulin / glucagon production by and release into the bloodstream.**
- **Negative feedback**
  Corrective mechanism **(OWTTE)** in the effector (target cells) brings the deviation in either direction back to its norm / set point

(d)(ii) Suggest a reason why glucose is considered as one of the most important metabolites. [1]
Universal respiratory substrate/ Brain only utilizes glucose/ Important precursor to generate intermediates that can lead to synthesis of amino acids and lipids

QUESTION 2

(a)(i) With reference to **Fig. X.1**, Identify stages **K**, **L**, **O** and **Q**. [2]

K: Prophase | **L**: Metaphase
O: Anaphase | **Q**: Cytokinesis

(ii) Describe the behaviour of chromosomes at stage **N** and **P**. [2]
At stage N, **sister chromatids separate**, **forming chromosomes**
chromosomes **move to opposite poles of the cell centromeres first** [® chromatids move to opposite poles of the cell (owtte)]
At stage P, chromosomes **reach opposite poles of the cell and start to uncoil** [® chromatids reach opposite poles of the cell and uncoil (owtte)]

(b) M checkpoint occurs during mitosis of the cell cycle. Explain the significance of this checkpoint. [2]
M checkpoint occurs in **metaphase, ensures** that **all chromosomes are aligned at the metaphase plate and kinetochore microtubules are attached to the centromeres of all chromosomes before proceeding to anaphase** and subsequently **exit the M phase of the cell cycle.** This **prevents chromosomes from being separated unequally into 2 daughter cells/ so that non-disjunction would not occur.** Cell cycle would also halt if there is damaged DNA.

(c)(i) With reference to Table 1, describe the differences in % distribution of cells at each stage. [2]
Majority of the cells **(87.3%)** are in **interphase stage/ 12.7%** of the cells are **undergoing cell division** and **lowest** percentage (**0.8%**) of cells found in **anaphase.** Approx equal distribution of the rest of the cells (**3 -4 %**) in **prophase, metaphase and telophase (name at least 2 stages).**

(ii) Using information from Tables 1 and 2, explain the effect of oncogene on cell division. [3]
Oncogene **increases the rate of cell proliferation** by **<u>permitting more cells to enter S phase of the cell cycle and subsequently mitosis</u>.** From table 1 and 2, there are number of cells in interphase decreases from 87.3% to 42.9% **/** any other comparison between other stages of cell division.

(iii) Suggest why it is important to use percentage values when comparing data from the normal and tumour cell cultures. [1]
because the **initial number of cells** in each culture may **not be the same** and hence it will **not possible to make a fair/valid/common comparison using absolute values** (OWTTE)

QUESTION 3

(a) Describe how binary fission in bacteria brings about genetic stability. [2]
asexual reproduction, where prior to binary fission, DNA is replicated via **semi-conservative replication** and the **genetic material is divided equally** to the **2** daughter cells ensuring **genetically-identical** daughter cells

(b) Describe how genetic variation is brought about in conjugation. [2]
Donor cell containing **F factor /** either as a **plasmid** (**F^+ cell**) or **integrated into the chromosome** (**Hfr cell**)**,** is capable of **forming a sex pilus / mating bridge**with a **recipient cell that lacks the F factor. DNA** from donor cell is **transferred from donor to recipient cell and is integrated into recipient strain via recombination**, hence bringing about genetic variation.

(c) Describe the trend observed in **Fig. 7.1**. [2]
Recombinants with **resistance to sodium azide are observed at highest frequency of 90%**OR **earliest at 10 min** into the experiment. Recombinants with the **ability to utilize lactose are observed at lowest frequency of 25%** OR **the last to be observed** at **25min** into the mating.

(d) Explain the relative positions of the genes seen in **Fig. 7.2**, using results from **Fig. 7.1**. [2]
Recombinants with resistance to sodium azide are **observed earliest** implying that the gene **is the first to be transferred** from Strain A to B. Hence it is **located nearest** to the origin of gene transfer. Subsequently, the order of recombinants observed are those with **resistance to T1, followed by ability to utilize glucose and finally ability to utilize lactose**. Thus **ability to utilize lactose is located furthest** from origin of gene transfer**. Recombinants with resistance to T1 are observed immediately** after recombinants with resistance to sodium azide, hence the **2 genes are located close to each other.**

(e)(i) State any difference there might be in the growth rate between the 2 tubes. [1]
No difference initially thereafter higher growth rate in the tube with lactose.

(ii) Explain your answer to **(e)(i)**. [2]

This is because at **high glucose concentration, cAMP concentration falls.** There is thus **no binding between cAMP and CAP / CRP. CAP / CRP** assumes an **inactive shape**, **detaching from the lac operon. Transcription of the lac operon** proceeds at only **a low level, even in the presence of lactose.** After **glucose is fully utilized, lac operon is switched on** and **growth rate in tube with lactose increases.**

QUESTION 4

(a)(i) Name one such chromatin protein. [1]

Histone(s) / Any of **histone H1, H2A, H2B, H3 and H4, Scaffolding proteins**, DNA methyltransferase
[R: endonucleases]

(ii) Explain how these proteins protect the gene concerned from DNase I digestion. [2]

DNA wound/wrapped around /interacts with histones. Proteins play a role in condensation of chromatin.
ref formation of **nucleosome** structure / heterochromatin
ref formation of solenoid
as **histones deacetylated** OR **acetate** group **removed from histone tail** / from **basic aa** on **histones**, increasing attraction betw basic histone and acidic DNA .
ref attraction between negatively charged (phosphate groups of) DNA and positively charged histones
ref positive charges of histones due to lysine residues (in histone tails)
gene DNA not exposed / accessible to DNase I enzyme / DNA double helix **cannot fit into active site of DNase I**

(b) With reference to the table,

(i) describe the differences in the levels of binding of the two types of cDNA probe for the developing RBC samples. [2]

Lower %tage binding of **globin cDNA probe** (quote value **25%**)
Higher %tage binding of **ovalbumin cDNA / probe** for developing RBC sample (quote value **93%**)

(ii) which gene is more actively transcribed in developing red blood cells? [1]

Globin gene

(iii) explain why the experimental result supports your answer to **(ii)**. [2]

Globin gene DNA **not wound around histones** / ref **histones within globin gene region acetylated / less tightly coiled** (around histones). Globin gene found within **euchromatin** / loosely coiled chromatin **whereas ovalbumin gene found within heterochromatin.**
Promoter region / gene **accessible to transcriptional factors** and **RNA polymerase II** / **transcriptional machinery** / RNA polymerase as shown by **larger extent of DNase I digestion** / resulting in globin gene DNA was **relatively less protected** against DNase I digestion / ref data before and after DNase I treatment, 94% vs 25% with **little remaining to hybridise with globin cDNA probe**.
Ovalbumin cDNA % binding almost unchanged at 93% / relate to reduced degradation of ovalbumin genes.
Ref **nucleosome re-modelling**

(c)(i) State the role of the β-globin promoter. [1]

Contains **RNA pol II binding site** for formation of **tcpnal initiation** complex / for initiation of tcpn. Binding of initiation complex proteins / TFIID controls/regulates transcription of β-globin gene. Controls efficiency of transcription/transcription initiation, a cis-acting regulatory sequence, promotes transcription.

(ii) With reference to figure X, explain how β-globin expression in the developing red blood cell may be regulated by regulating levels of NF-E4 within the nucleus. [2]

NF-E4 binds to β-globin promoter / NF-E4 binding site within β-globin promoter via DNA-binding domain. Possibly acts as (basal) TF / activator protein / enhancer-binding protein recruiting basal TFs, increases rate of formation of stable / stability of transcriptional initiation complex & stabilises RNA polymerase (II) binding to TATA box. This increases rate of promoter clearance. **Upregulation/increased levels of nuclear NF-E4 leads to increased rate of transcription of β-globin gene**
Ref to <u>either</u> up- or down-regulation
Ref to terms for prokaryotic system of controls e.g. operon, inducer

QUESTION 5

(a) Define what is meant by a pure-bred plant. [1]

A plant that is **homozygous at all gene loci** OR produces only **offspring** of the **same variety** when it is **self-pollinated.**

(b) Using suitable symbols, draw a genetic diagram to explain the results of the test cross. [5]
Let **A** represent the **dominant allele** for **green leaves**, **a** represent the **recessive allele** for **mottled leaves.**
B represent the **dominant allele** for **smooth fruit**, **b** represent the **recessive allele** for **hairy fruit.**

F_1 test cross Green leaves, smooth fruit x Mottled leaves, hairy fruit

$\frac{\mathbf{A\ b}}{\mathbf{a\ B}}$ $\frac{\mathbf{a\ b}}{\mathbf{a\ b}}$

Gametes A b a B A B a b a b

Random Fertilization

		Female Gametes			
		A b	a B	A B	a b
Male Gametes	a b	$\frac{\mathbf{A\ b}}{\mathbf{a\ b}}$	$\frac{\mathbf{a\ B}}{\mathbf{a\ b}}$	$\frac{\mathbf{A\ B}}{\mathbf{a\ b}}$	$\frac{\mathbf{a\ b}}{\mathbf{a\ b}}$

Phenotypes	Green leaves, hairy fruit 46	Mottled leaves, smooth fruit 40	Green leaves, smooth fruit 8	Mottled leaves, hairy fruit 6
	Parental		**Recombinants**	

(c) Explain the deviation of the observed test cross results from the expected dihybrid test cross ratio. [2]
Expected dihybrid test cross ratio is **1:1:1:1** while **observed test cross results do not have a definite ratio** which indicates that the **2 genes** for leave colour and fruit texture are **incompletely linked.** As such, there is **crossing over** between the **2 linked genes** (in only some of the cells) thus the observed offspring of test cross are **majority of parental combinations** and **minority of recombinants.**

(d)(i) This second scientist claimed that the observed test cross ratio of offspring can be simplified to 1:1. Perform the chi-squared (χ^2) test to verify his claim. [2]

$\chi^2_{\text{calculated}}$ **= [(52-48)^2 + (44-48)^2]/48 = 0.67**

At p = 0.05, degree of freedom = 1, $\chi^2_{\text{critical (0.05,1)}}$ **= <u>3.84</u>**

Since $\chi^2_{\text{calculated}} < \chi^2_{\text{critical}}$, there is **<u>no significant difference</u>** between **<u>observed test cross ratio and expected 1:1 ratio</u>.** Hence, the scientist's **claim is valid.**

(ii) Suggest why this second scientist made this claim. [1]
The **2 genes** for leave colour and fruit texture are so **closely linked** OR **crossing over** between the 2 genes for leave colour and fruit texture is so **rare** such that they are **inherited as a unit / do not show independent assortment** during **meiosis**

QUESTION 6
(a)(i) State the level of protein structure shown in **Fig. 6.2**. [1]
Tertiary

(ii) With reference to **Fig. 6.1** and **6.2**, explain how the structure of the rhodopsin receptor allows it to carry out its function. [2]
Consists of **seven α helices forming a transmembrane protein, thus allowing the transmission of the extracellular into an intracellular signal.** Formation of **specific 3D conformation complementary to the specific ligand,** allowing for formation of **ligand receptor complex to activate** downstream **transduction pathway.**

(b)(i) State any two steps in **Fig 6.3** which result in amplification effect of this signalling pathway. [1]
A/ B/ C (any 2)

(ii) Explain the significance of the signal amplification effect. [2]
Multiplicity of steps **which results in a** strengthened / **an** enhanced response to a small stimulus. Provides more opportunities for **coordination/ control/ regulation** to generate **large-scale coordinated response/ multiple responses.**

(c) Describe the trends observed in **Fig 6.4** for **E1** and **E2**. [2]
As **stimulus intensity increases, the frequency of action potential increases.** In **E2, a weaker stimulus** results **in 7 action potentials being produced.** Compared to **E3,** whereby a **stronger stimulus** results in **9 action potentials** for the same time period, **peak / amplitude** of **each action potential is the same, at about +15 mV.**

(d)(i) State what do you understand by the term threshold potential. [1]
refers to the **potential in which a membrane must be depolarised / reached for an action potential to be triggered,** has a **value of about –55 mV.**

(ii) With reference to the results shown in **Fig 6.4** and **Table 6.1**, explain how stimulus intensity affects the extent of muscular contraction. [3]
In **E1,** a **weak stimulus** that produces a **subthreshold graded potential / graded potential** that is **below threshold will not trigger an action potential. No neurotransmitters** will be **released** into the synaptic cleft **to cause twitches.** A **strong stimulus** causes the **membrane potential to reach threshold, triggering a wave of action potentials** which will result in the **release of neurotransmitters to cause twitches.** in **E3, 11 twitches are produced per second as compared to only 7 in E2** since a **stronger stimulus is applied to the neurone in E3,** resulting in a **greater release of neurotransmitter.**

QUESTION 7

(a)(i) State 1 difference between traditional classification and phylogenetic systematics. [1]

Traditional classification	Phylogenetic classification
• uses both **shared ancestral** & **derived** characters	• use **only shared derived** features
• **some descendants** of common ancestor **not** included	• **all common descendants** of group included

(ii) **P, Q** and **R** show 3 types of grouping. Identify the groupings. [3]

P	Paraphyletic
Q	polyphyletic
R	Monophyletic

(ii) Explain how these types of groupings are derived. [2]
R is made up of a group of organisms **containing all the descendants** of that **most recent common ancestor**. P is a monophyletic group that includes the **ancestor & some but not all** of its descendants. Q is made up of groups that **do not include the common ancestor**.

(b)(i) Draw a box to include the entire region that most likely codes for an important structure in the mitochondrion. [1]

```
England         AAAGCAGAAGAAATAGATGCAAAGAAGAAGAGTTCAA
Japan           --GCAT--------------C------------AATT
South Africa    -------C-----------------TA----------
India           GG-CG--C------------GGG-GG-------CGGG
China           TTT----T------------TT------T----GGTT
United States   -------C------------T----------------
Philippines     TTT----T------------TC------T----GGTT
New Zealand     -------C---------------C-TA----------
Russia          --GCAT--------------CC-----------A-TT
Yemen           ------------------------------C------
```

(ii) Explain your answer for **(b)(i)**. [2]
By comparing homologous sections where **DNA sequence is highly conserved/ identical**. This is due to **absence of nucleotide substitution** as the region codes for **an important part of a protein**/ protein that contributes to **normal functioning of mitochondrion**.

(iii) With respect to **Fig. 7.2**, identify the country of the subspecies which is most closely related to the mouse found in England. Give a reason for your answer. [2]
Yemen, since there is only a change in one nucleotide base where **guanine** is **substituted by cytosine**.

QUESTION 8

(a) Describe the structural features of a named membranous organelle that is involved in ATP synthesis in photosynthesis. [6]

1. **chloroplast**
2. **large / long / elongated / lens-shaped / 5 – 10 μm in length**
3. enclosed by a **<u>double membrane</u> / <u>inner and outer membrane</u>**
4. **inner membrane is <u>selectively permeable</u>** containing protein channels
5. **outer membrane is <u>permeable</u>** to most ions and molecules
6. contain **gel-like aqueous matrix known as <u>stroma</u>**
7. **starch grains / ribosomes / circular DNA / enzymes of Calvin cycle present**
8. flattened, disc-like sacs known as **<u>thylakoids</u> suspended in stroma**
9. arranged in **stacks known as <u>grana</u>**
10. which are **joined by <u>intergranal lamellae</u>**
11. **thylakoid membrane**
12. contains **photosystems**

12a. **photosynthetic pigments** e.g. chlorophyll and carotenoids are present

13. **electron carriers** / **electron transport chains** are also embedded in the membrane
14. **$NADP^+$ reductase** present
15. **ATP synthase complex / stalked particles** also present
16. **fluid-filled interior space** known as **<u>thylakoid lumen / space</u>**

(b) Compare the two main pathways in which an excited electron can undergo during the light-dependent stage of photosynthesis. [6]

Differences:

Features	***non-cyclic photophosphorylation***	***cyclic photophosphorylation***
Photosystems involved	involves both **<u>PSII</u>** and **<u>PSI</u>**	involves **<u>only PSI</u>**
First electron donor (source of electrons)	from **<u>water</u>, photolysis of water occurs**	from **<u>P700 in PSI</u>,** no need water s.t. **photolysis of water does not occur**
Electron transport	involves **plastoquinone, cytochrome complex, plastocyanin** and **ferredoxin.** Also involves **two electron transport chains.**	involves **ferredoxin**, **cytochrome complex** and **plastocyanin only &** involves **only one electron transport chain**
Pathway of electron	**<u>non-cyclic</u>**, electron does not return to same molecule	**<u>cyclic</u>**, electron returns to same molecule
Final electron acceptor	**<u>$NADP^+$</u>**	**<u>P700 in PSI</u>**
Generation of proton gradient	proton gradient generated by H^+ released during **photolysis of water** , and **H^+ that are actively pumped** from the stroma, across the thylakoid membrane, **into the thylakoid lumen by energy released during electron flow.**	proton gradient is generated by **H^+ that are actively pumped** from the stroma, across the thylakoid membrane, **into the thylakoid lumen by energy released during electron flow**
Products	**<u>ATP</u>, <u>NADPH</u> & by-product O_2 produced**	**<u>ATP only</u>, no O_2 produced**

Similarities:

- both pathways **occur at the thylakoid membrane**
- both pathways involve **PSI**
- both involve the **synthesis of ATP**
- both pathways uses **energy released during electron flow** to actively **pump H^+** from the stroma, across the thylakoid membrane, **into the thylakoid lumen to produce a proton gradient** which drives ATP synthesis **/** involves **chemiosmosis**

(c) With reference to the term 'limiting factor', discuss how two named factors have effects on photosynthesis. [8]

1. rate of photosynthesis is **limited by the slowest reaction in the series**
2. and when the rate of photosynthesis is **limited by more than one factors**
3. the **rate is limited by the factor which is in the shortest supply**
4. it **determines the rate of reaction**

5. **light intensity**
6. directly affects **light-dependent stage** of photosynthesis
7. at **low light intensities**, **light intensity is the limiting factor**
8. **rate of photosynthesis increases proportionally with increasing light intensity**
9. at **high light intensities**, **rate of photosynthesis decreases**
10. **some other factor becomes the limiting factor**
11. **further increase in light intensity** results in the **light saturation point being reached**
12. beyond which **further increase in light intensity will have no effect on the rate of photosynthesis**
13. **light intensity is no longer a limiting factor**
14. **rate of photosynthesis is maximum at this point**

15. **carbon dioxide**
16. a **major limiting factor** since atmospheric CO_2 concentration is low
17. CO_2 is **needed in the light-dependent stage / Calvin cycle**
18. it is **reduced to carbohydrate / sugar**
19. it is **fixed by combining with ribulose-1,5-bisphosphate (RuBP)**
20. **increase in CO_2 concentration results in a proportional increase in the rate of photosynthesis**

21. **temperature**
22. affects **mainly light-independent stage** and, to a certain extent, the light-dependent stage
23. are **enzyme-controlled**
24. **increase in temperature, increases rate of photosynthesis**
25. **rate of photosynthesis doubles for each rise of 10°C** until optimum temperature is reached
26. **rate of photosynthesis decreases at higher temperature**
27. as **enzymes start to denature**

28. **oxygen**
29. **competes with CO_2**
30. for the **active site of the enzyme RuBP carboxylase oxygenase / Rubisco**
31. during the **light-independent stage / Calvin cycle**
32. **high concentration of oxygen decreases the rate of photosynthesis**

QUESTION 9

(a) Describe the structural features of **two** named nucleic acids involved in protein synthesis. [6]

DNA/ deoxyribonucleic acid/ deoxyribose nucleic acid

1a. There are **four types** of **deoxyribonucleotides/ deoxyribose nucleotides**
1b. composed of **Adenine**, **thymine**, **cytosine** and **guanine**
3. Nucleotides are linked together by **phosphodiester bond**
4. **DNA molecule is double-stranded**
5. 2 strands held together by **hydrogen bonds** between **complementary bases**/ **2 hydrogen bonds** between **A and T** and **3 hydrogen bonds** between **C and G**
6. Bases are found between the 2 sugar-phosphate backbone
7. Bases are **stacked** above one another with a distance of **0.34 nm**
8. There is **hydrophobic interaction** between the stacked bases, which stabilizes the molecule
9. The two polynucleotide strands are **anti-parallel** / One polynucleotide strand runs in the **3' → 5' direction** while the other polynucleotide strand runs in the **5' → 3' direction**
10. **DNA molecule adopts a helical structure**
11. There are **10 nucleotides per turn** of the helix, with a length of **3.4 nm**
12. The **sugar-phosphate backbone** is found on the **outside of the helix**
13. the **hydrophobic nitrogenous bases** are found in the **core of the helix**
14. **Major** and **Minor grooves** are found on the DNA double helix

mRNA

1a. **Four types** of **ribonucleotides**
1b. composed of **adenine**, **uracil**, **cytosine** and **guanine**
2. Held together by **phosphodiester bond**
3. RNA molecule is **single-stranded**
3a. RNA molecule has a 3' hydroxyl end and a 5' phosphate end
4. Has a **5' guanine cap**
5. Has **3' poly-A tail**
6. **exons are joined togther** in the coding region
6a. contains a coding region
7. Has 5' and 3' untranslated region
8. Forms fairly random coil

tRNA
1a. **Four types** of **ribonucleotides**
1b. composed of **adenine**, **uracil**, **cytosine** and **guanine**
2. Held together by **phosphodiester bond**
3. RNA molecule is **single-stranded**
4. Its **2D structure** is in the form of a **cloverleaf**.
4a. formed by complementary base-pairing
5. **one of the loops contains the anticodon**
6. Its **3D structure** is a **compact L-shaped structure**
6a. maintained by hydrogen bonds
7. Has a **3' CCA stem** for the **attachment of amino acid**

(b) Compare the two processes that have to take place for protein synthesis. [8]

Similarity
(1) Both processes **require a template**.
(2) Both processes are **catalyzed by enzymes**.
(3) Both processes involve **complementary base-pairing**.
Differences

Feature	*Transcription*	*Translation*
Location	**In nucleus**	**ribosomes**
Involvement of ribosome	**not involved**	**Ribosome is the binding site for mRNA** and **amino-acid tRNA complex**.
Template	**DNA template/ coding strand**	**mRNA**
Enzyme	**RNA polymerase catalyses formation of phosphodiester bond** between adjacent ribonucleotides	**peptidyl transferase** catalyzes the **formation of peptide bond**.
Bond between basic units.	**Bond is formed** between the **phosphate group of one nucleotide** and **carbon atom 3 of ribose** of the next nucleotide.	**Bond is formed** between **carboxyl group of one amino acid** and the **amino group of the next amino acid**.
reading of genetic message	**RNA polymerase moves along DNA template/ coding strand.**	**Ribosome moves along mRNA.**
Involvement of tRNA	**not involved**.	**tRNA carries amino acids** to ribosomes.
raw material(s)	**free ribonucleotides**	**amino acids attached to tRNA**
product(s)	**mRNA**, **rRNA** and **tRNA**	**polypeptide**
Fate of product(s)	Products **exit nucleus** and **migrate to the cytoplasm**	**polypeptide chain remain in the cytoplasm** or **secreted out of the cell**

(c) Discuss two named factors that would affect the rate of protein synthesis. [6]

Named factor 1: Stability of mRNA
mRNA that is **more stable** enables **translation to proceed for a longer period of time**
More proteins can thus be produced per unit time

Named factor 2: Presence of internal ribosome entry site (IRES)
This allows translation to take place even when the **cap binding complex, eIF-4E** is not able to **bind to the 5' cap increasing the rate of translation**

Named factor 3: Presence of upstream open reading frames (uORFs)
uORFs **decrease translation of downstream gene**
Ribosome **translate the uORF** and **dissociate from the mRNA before it reaches the coding sequence**

Named factor 4: Presence of translational repressors
Translational repressors **decrease the rate of translation** by **preventing communication between 5' cap and 3' tail** which is **required for efficient translation**

QUESTION 1

(a) In the space provided in **Fig. 1.2**, indicate with

(i) the symbols + and -, the relative positions of the electrodes, and

(ii) an **arrow,** the direction of migration of restriction fragments during gel electrophoresis. [2]

Negative on top & positive below. Arrow points downwards.

(b) State what is meant when a certain locus is described as 'polymorphic'. [1]

Multiple alleles occur at the locus (within the population). Variation in nucleotide base sequence at the locus at **frequency** that is **higher than that can be accounted for by mutation**. *[R: definition wrt RFLPs only]*

(c) Describe how RFLP may be used to map the chromosomal locus of a dominant mutant allele. [3]

Linkage analysis where **RFLP locus acts as genetic marker**. **Co-inheritance / cosegregation** of **RFLP allele** of which chromosomal locus is already known **with dominant trait / phenotype** within **affected family**. Indicates RFLP locus **closely linked** to dominant mutant allele. If certain RFLP alleles not co-inherited, can rule out which chromosomal loci are not closely linked to the allele.

(d) Based on the results shown in **Fig. 1.2**,

(i) state the genotypes of individuals **I-2** and **II-3** with respect to the **P** and **Q** loci. [2]

Individual **I-2** :	**P2P3**	**Q1Q1**
Individual **II-3** :	**P1P3**	**Q1Q2**

(ii) state on which chromosome the disease gene is located. [1]

Chromosome 3

(e) Individual **II-6** marries a woman with the genotype **P1P2** and they have a child who is homozygous for allele **P1**. The father suspects the child was illegitimate.

(i) Explain the genetic basis for this suspicion. [2]

Child **inherits** one **P1 allele from mother** but individual II-6, being of **genotype P2P3 does not have P1** allele / cannot possibly contribute P1 allele. Hence child could have been **fathered by another man carrying P1 allele**. Legitimate child should be of either P1P2, P2P3, P2P2 or P1P3 genotype / cannot possibly have P1P1 genotype.

Ref idea that child must inherit one chromosome / one P allele from each parent;

(ii) If the child is not illegitimate, suggest one possible explanation how the child could have acquired the **P1P1** genotype. [2]

Point mutation / logical named type of mutation **within germ-line** cells in individual II-6 *[R: mutation occurred in child]* within **middle XhoI** site **of either P2 or P3 allele**. Hence this **restriction site destroyed / eliminated** resulting in P2 or P3 allele converted to P1 allele so **child inherits P1 allele from II-6** instead.

(iii) Suggest and explain what further test(s) may be used to confirm the suspicion. [2]

DNA fingerprinting / **PCR-STR** using **multiple/another VNTR/STR/RFLP loci/locus**. Analysis of another RFLP locus e.g. locus Q. If II-6 is the father, child would have inherited paternal alleles of II-6 at these other loci / (clear explanation of interpretation of results i.e. for these other loci, child should have RFLP band pattern that corresponds to inheritance of alleles present in mother and II-6 / present in biological parents / OWTTE). Spontaneous mutations being rare, hence not likely to occur at other loci in germ-line of II-6.

QUESTION 2

(a) Describe how the SGP serves to increase knowledge of the biology of Atlantic salmon. [2]

To generate genomic maps, to identify and analyse genes and their regulatory sequences, as well as study **patterns of inheritance of genes,** and their **evolutionary relationships with other organisms.**

(b) With reference to **Fig. 1.1**, describe the features of the MCS. [2]

Consists of a **variety of restriction enzyme sites** s.a. ***BssH* II, *Sac* I** and ***Kpn* I** that **occur only once / unique** in the vector. Found within selection marker lacZ for **cloning of gene of interest** into vector**.**

(c) (i) It was found that the recombinant plasmid with sGH gene inserted yielded no polypeptide. State and explain a reason for this. [2]

sGH gene inserted in the **wrong orientation** though **a protein is expressed under control of T3 promoter. Sequence of gene is reversed** and **3D conformation of protein is different** from original**.** OR Since ***Sac* I restriction site is within / near T3 promoter**, sGH gene inserted **disrupts promoter sequence**. This results in **no expression** of gene & thus **no polypeptide** is produced.

(ii) Suggest a solution for the problem in **(c) (i)**. [1]
Use **2 restriction enzymes** to clone**/** use **restriction site which is away from T3 promoter** to clone**/ construct transgene with promoter in front of sGH gene**

(d) (i) Describe the results shown in **Fig. 1.2**. [2]
Domestic GH-treated reached **maximum weight of 130g** while **Domestic-control** reached **maximum weight of 90g**/ **Domestic GH-treated** had **increase in weight gain of 120g** while **Domestic-control** had **increase in weight gain of 80g**/ **Wild GH-treated** reached **maximum weight of 65g** while **Wild-control** reached **maximum weight of 15g**/ **Domestic GH-treated** had **increase in weight gain of 55g** while **Domestic-control** had **increase in weight gain of 5g.**
Domestic salmons had higher weight gains as compared to wild salmons. **Domestic GH-treated** reached **maximum weight of 130g** while **Wild GH-treated** reached **maximum weight of 65g/ Domestic control** reached **maximum weight of 90g** whil**e Wild-control reached maximum weight of 15g.**

(ii) Suggest a reason for the difference in weight gain between domestic-GH treated and Wild-GH treated fish. [1]
Other factors eg. **Environment, nutrients**, also play a role in weight gain**.**

(iii) Describe the results seen in **Fig. 1.3**. [1]
sGH extracted on 11/5 is of lower molecular weight than those extracted later**.**

(iv) With reference to both **Fig. 1.2** and **Fig. 1.3**, suggest the process sGH undergoes before exerting its effect on weight gain. [1]
On 11/5, sGH is of higher molecular weight, hence it is **first synthesized as a pre-hormone.** It then **undergoes post-translational modification** where **some peptide sequences are cleaved off** to form the **mature / functional hormone.** Hence accounting for **sGH of lower molecular weight** and accounting for **increased weight gain** as seen **in later dates.**

(e) (i) With reference to **Fig. 1.5**, state the one selection pressure faced by transgenic salmon. [1]
mating advantage / offspring vialibity

(ii) Using your answer to **(e) (i)**, explain how the named selection pressure may or may not lead to salmon extinction. [2]
(Offspring viability) From **0.7 to 1.0 of offspring viability**, **increase in offspring viability** leads to **increase in time to extinction** from **35 generations to more than 150 generations.** From **0.4 to 0.7 of offspring viability**, **increase in offspring viability** leads to **decrease in time to extinction** from **125 generations to more than 35 generations.**
(Mating advantage) **As mating advantage decreases, the minimum time to extinction increases for example,** minimum time to extinction for a **mating advantage of 4.7** is about **35 generations** as compared to minimum time to extinction of **more than 150 generation**s for a **mating advantage of 1.4.**

QUESTION 3
With reference to **Fig. 3.1**,
(a)(i) name the process in which the blue cells were formed from the isolated cells. [1]
Mitosis

(ii) are the blue cells stem cells? Give a reason for your answer. [2]
Yes as they are **capable of self-renew/ giving rise to other cell types.**

(iii) given that the green cell is a stem cell, explain why green cells can give rise to differently coloured cells. [2]
Green cell is **multipotent.** As such, they can **give rise to more than one cell type.** They can be **influenced** by the **presence of internal** and **external signals** & are **capable of asymmetric division** which involves **unequal distribution of cytoplasmic determinants** as well as **some genes being switched on/ of. Green cell undergoes differentiation** when **in contact with neighbouring cells** and **responds to chemical signals.**

With reference to **Fig. 3.2**,
(b)(i) name the type of gene therapy involved. [1]
Ex vivo OR **Somatic cell** gene therapy

(ii) suggest why the cells in **Steps 1** to **3** were grown on other cells when cultured. [1]
Give the mESCs a sticky surface to which they can attach OR **Provide / release nutrients into culture medium.**

(iii) give a reason why cells with markers consistent with pancreatic islet cells from **Step 2** were selected for further differentiation in **Step 3**. [1]

Presence of markers consistent with islet cells indicate that the cells are **similar to the pancreas cells** and are able to **detect the changes in blood glucose concentration.** There is evidence that the transgene is **successfully incorporated** into the cells & is **being expressed.**

(c)(i) When the pancreatic-like cells were placed in the diabetic mouse, explain briefly why it was necessary for vascularization to take place. [2]

Vascularization ensured that the **changes in the blood glucose concentration of the mouse can be detected** by the pancreatic-like cells **through the circulatory system** (OWTTE)**.** The **insulin** produced can be **released directly into the bloodstream/ circulatory system to travel to the target effector cells/ tissues/ organs** to bring about a **lowering to the blood glucose concentration of the mouse to set-point.** The pancreatic-like cells **received nutrients and oxygen** for the **survival of the cells**.

(ii) Outline the role of glucose-sensitive promoter in this method of gene therapy for mice. [1]

Insulin is a **peptide hormone.** Glucose-sensitive promoter **switches on / upregulate transcription / gene expression** to produce mRNA / insulin. **As blood glucose rises insulin production increases, vice versa.**

(iii) With reference to **Fig. 3.3**, account for the results obtained. [2]

Following transplantation, the daily mean blood glucose concentration of the diabetic mouse **dropped / decreased** from **21 mg/dm^3 (day 0) to 7 mg/dm^3 (day 110)** and **maintained** approximately **at set point** of **7 mg/dm^3 from day 110 onwards to day 150.** This indicates that the mouse was able to **produce insulin to <u>regulate the daily mean</u> blood glucose concentration** over a **substantially long duration of time** (OWTTE).

(iii) Discuss one possible **benefit** and **problem** of using this gene therapy in the treatment of diabetes in humans. [2]

BENEFIT:
- **Avoids injections / pain of injections**
- **Mimics normal pancreatic behaviour**
- **More stable homeostasis / reduced highs and lows in blood sugar**
- **Less chance of hypoglycaemia / hyperglycaemia**
- **Less restriction on lifestyle**
- **No need to measure blood sugar**

PROBLEM:
- **Transplant rejection**
- **Cells could lodge elsewhere and not at the target site**
- **Transgene may have unforeseen effect**
- ***AVP* e.g. rat data may not be applicable to humans**

QUESTION 4

(a) Using a tomato leaf, describe in detail how protoplast culture can be used for the micropropagation of the plant. [8]

1. **tomato leaf is disinfected** with dilute bleach solution
2. **cut** / **dissect tomato leaf into pieces**
3. **soak** leaf parts in **solution containing microbial enzymes / cellulase**
4. to **digest cell wall of plant cells**
5. **protoplasts can be obtained from cells in suspension / directly from mesophyll leaf cells**

6. **<u>establishment of protoplast cells in culture</u>**
7. protoplast cells **transferred to agar medium with nutrients and growth regulators /** ref. made to **named nutrients and growth regulators e.g. auxin and cytokinin**
8. tissue culture **grown in special, sealed transparent / GA7 containers**
9. **in vitro**
10. **culture medium sterilized**
11. **addition of osmoticum / mannitol / glucose / sucrose**
12. to **maintain osmotic pressure** to **prevent disruption of membranes**

13. **<u>multiplication of microshoots</u>**
14. from **callus**
15. **increase cytokinin concentration** in culture medium / **high cytokinin : low auxin ratio**
16. after microshoots have reached an appropriate length, they can be **subcultured to new multiplication media**

17. <u>**microcuttings**</u> obtained from microshoots <u>**undergo root formation**</u>**;**

<u>*EITHER*</u>

18. **in vitro rooting**
19. whereby **microcuttings are rooted in a test-tube containing root-inducing medium**
20. **concentration of auxin in the medium is high** while **concentration of cytokinin is kept low / high auxin : cytokinin ratio**

<u>*OR*</u>

21. **ex vitro rooting**
22. **microcuttings treated with auxin**
23. inserted in **soil-less greenhouse rooting medium**
24. **placing them under mist / high humidity conditions**

25. <u>**acclimatization**</u> of the **tomato plantlets**
26. **to the outdoor environment**

(b) Explain why farmers still use traditional methods to clone their crops despite the availability of tissue culture technologies. [4]

1. not feasible to clone crops when they are required in **very large numbers** using tissue culture technologies as it can be **labour-intensive**
2. all **clones** are **genetically identical** so any change in environmental conditions / appearance of a new disease could have **devastating consequences / lead to extinction** if the plants are not resistant or cannot adapt
3. equipment required for tissue culture technologies are **expensive / costly / high in cost** individual **plants** must have **high market value** and thus **more relevant for ornamental plants than crop plants** OR workers need to be trained and **farmers** may **not** be **able to afford** them
4. **crops** are **grown in controlled and optimized conditions** in tissue culture **transition** of **crop from medium to soil** may pose a **problem / success rate of transplanted crops is low**
5. tissue culture techniques must be carried out in **sterile / aseptic conditions** thus **requires highly-trained staff** and posing **severe constraints on working practice**
6. tissue culture technologies are **relatively new unforeseen problems**, e.g. genetic mutation **may arise**

(c) Like plant transgenesis, the human genome project is a landmark scientific breakthrough. Discuss the benefits and associated ethical implications of both breakthroughs. [8]

<u>**Human Genome Project**</u>

Benefits	**Ethical Implications**
- **Pharmacogenomics** / knowing about which genes affect a person's response to a drug / genetic differences affect the way we react to a drug**.** **Possibility of tailoring drugs** prescribed to fit patient's pharmacogenomic profile for **greater efficacy** / **avoiding dangerous side effects**.	- Most ***effective drug*** may ***not be affordable*** for patient**.** Such ***testing*** may not be ***affordable & psychological impact*** of knowing treatment is available but not affordable**.**
- **Risk assessment** for certain diseases**. Knowledge of genetic variations** can help **prevent / treat these diseases.**	- ***Risk profiles*** for diseases may be ***accessible to insurers, employers, schools etc / individual's privary infringed.*** May be used in ***discriminatory manner***/ ***stigmatization***/ ***social risks***. - ***Psychological impact*** due to an individual's ***genetic differences*****.** Doctors, other health service providers, patients and the public must be ***educated*** to make ***informed choices,*** and ***be aware of scientific capabilities*** and ***limitations*****.** - ***Standards*** and ***quality-control measures*** in ***testing procedures*** must be ***enforced*** to ***ensure reliability of tests*****.** Proper ***counselling*** must be done in conjunction with risk profiling and where there is ***no treatment*** for certain diseases**.** - ***Reproductive issues*** / ***decision making*** / ***rights*** and ***potentially controversial procedures*** e.g. eugenics, fetal genetic testing and abortion**.**

- Insights via **comparative genomics** / comparing human genome sequence and the genome sequence of model organisms / identification of genetic similarities and differences to model organisms**.** - Leads to **gene discovery** / follow **disease development** / study **disease treatment** / elucidating **evolutionary history** and **relationships** between genes, genome and species**.**	- Infringement of ***animal rights.*** Also, ***animal disease models*** may ***not accurately reflect disease in humans*****.**
- **Large scale studies** on genetically homogeneous populations to **identify alleles linked to certain disease** conditions**.**	- ***Uncertainties*** associated with ***gene tests*****.** - ***Genetic information of individuals*** becoming ***private intellectual property*** of biotechnology companies**.** ***Commercialization of related technologies*** derived from such research, but possibly ***no reciprocal direct benefit for individuals contributing*** their genetic information to the research**.**

Plant Transgenesis

Benefits	Ethical Implications
- May help to **alleviate nutritional deficiencies**/ **reduce malnutrition** in developing countries**. Reduces the incidence** of **impairment, infections** and **diseases** by **increasing the nutritional value** of crop plants**.** - **Sustainable** and **cost-effective alternative** to supplements**. Cuts down** on **heavy reliance** on **chemical insecticides,** thus **reduces cost, labour** & **environmental damage.** - GM plants with traits **leading to harvest yields** that are **much higher** than those achieved with cultivation of conventional crop lines**.** - **Economic losses** due to pest infestations are **greatly reduced. Efficient agricultural production** contributes towards **reducing costs in food-production sector.** - GM plants are **target-specific** and **very effective** in terms of their mode of action**.**	- Some individuals may find ***genetic manipulation of plants unacceptable*****.** To them, if ***humans have rights, so do other living organisms*** / it is ***inappropriate to tamper with nature*** by mixing genes among species, i.e. we ***should not be "playing God"*****.** - Some ***companies*** **may** want to ***patent transgenic plants*****.** Issue of ***biopiracy*** / ***companies*** need to ***protect*** their often lengthy and expensive ***research programs*****.** ***Patenting plants*** is ***unethical*** and it ***reduces them to the level of objects*****.** - ***Not all risks may have been evaluated sufficiently*** / inadequate risks assessment / not possible to anticipate all potential risks and consequences**.** It is ***morally wrong*** for companies to ***market*** their ***products without knowing for sure their potential health risks*** / side-effects**.** - To some, creation of transgenic plants ***will not solve the demand for food in Third World countries***, but will instead result in an ***increase*** in ***their dependence*** on rich developed countries**.** ***Richer nations*** thus ***benefit*** at the ***expense of poorer nations*** / ***world food production*** becomes ***dominated by a small number of large companies*** / ***farmers*** may be ***forced to buy fresh supply of seeds*** each year**.** - ***Religious concerns*** or ***dietary restrictions*** since animal genes may be inserted into the genome of plants**.** A need to be ***transparent*** since **consumers** have the ***right to information*** with regards to GMOs**.** ***Food labelling*** is not mandatory in some countries / ***regulations*** against mixing of GM foods with non-GM products are not ***strictly enforced*****.**